



PDF Download
3477495.3531867.pdf
18 January 2026
Total Citations: 37
Total Downloads: 1358

 Latest updates: <https://dl.acm.org/doi/10.1145/3477495.3531867>

RESEARCH-ARTICLE

Multimodal Entity Linking with Gated Hierarchical Fusion and Contrastive Training

PENG WANG, Southeast University, Nanjing, Jiangsu, China

JIANGHENG WU, Southeast University, Nanjing, Jiangsu, China

XIAOHANG CHEN, Southeast University, Nanjing, Jiangsu, China

Open Access Support provided by:

Southeast University

Published: 06 July 2022

[Citation in BibTeX format](#)

SIGIR '22: The 45th International ACM
SIGIR Conference on Research and
Development in Information Retrieval
July 11 - 15, 2022
Madrid, Spain

Conference Sponsors:
SIGIR

Multimodal Entity Linking with Gated Hierarchical Fusion and Contrastive Training

Peng Wang*
Southeast University
Nanjing, Jiangsu, China
pwang@seu.edu.cn

Jiangheng Wu
Southeast University
Nanjing, Jiangsu, China
220171694@seu.edu.cn

Xiaohang Chen
Southeast University
Nanjing, Jiangsu, China
220191804@seu.edu.cn

ABSTRACT

Previous entity linking methods in knowledge graphs (KGs) mostly link the textual mentions to corresponding entities. However, they have deficiencies in processing numerous multimodal data, when the text is too short to provide enough context. Consequently, we conceive the idea of introducing valuable information of other modalities, and propose a novel multimodal entity linking method with gated hierarchical multimodal fusion and contrastive training (GHMFC). Firstly, in order to discover the fine-grained inter-modal correlations, GHMFC extracts the hierarchical features of text and visual co-attention through the multi-modal co-attention mechanism: textual-guided visual attention and visual-guided textual attention. The former attention obtains weighted visual features under the guidance of textual information. In contrast, the latter attention produces weighted textual features under the guidance of visual information. Afterwards, gated fusion is used to evaluate the importance of hierarchical features of different modalities and integrate them into the final multimodal representations of mentions. Subsequently, contrastive training with two types of contrastive losses is designed to learn more generic multimodal features and reduce noise. Finally, the linking entities are selected by calculating the cosine similarity between representations of mentions and entities in KGs. To evaluate the proposed method, this paper releases two new open multimodal entity linking datasets: WikiMEL and Richpedia-MEL. Experimental results demonstrate that GHMFC can learn meaningful multimodal representation and significantly outperforms most of the baseline methods.

CCS CONCEPTS

• **Information systems** → **Multimedia and multimodal retrieval**.

KEYWORDS

Entity linking, Multimodal fusion, Contrastive training, Knowledge graph

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

[SIGIR '22] *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, July 11–15, 2022, Madrid, Spain.

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-8732-3/22/07...\$15.00

<https://doi.org/10.1145/3477495.3531867>

ACM Reference Format:

Peng Wang, Jiangheng Wu, and Xiaohang Chen. 2022. Multimodal Entity Linking with Gated Hierarchical Fusion and Contrastive Training. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '22)*, July 11–15, 2022, Madrid, Spain. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3477495.3531867>

1 INTRODUCTION

Entity linking (EL) is the task of aligning mentions to corresponding entities in knowledge graphs (KGs) [35]. It is crucial in many information retrieval applications, such as information extraction [20], question answering [24, 34, 46], and Web search query [4, 17]. The ambiguity of mentions poses a great challenge to EL. Previous approaches mainly disambiguate mentions via context resolution [6, 12, 20] and have been proved effective when the textual information is rich. However, it is still challenging to link mentions from short and coarse text in real-world multimodal data. For example, on social media, users tend to post brief text attached with some multimodal information, such as images, sounds and videos. In Figure 1, a short tweet only contains few words, "*Black panther, you will always be the pride of Wakanda. Rest in peace*", with a film poster to mourn for the actor *Chadwick Boseman*. If only according to the text, it is difficult for existing EL models to judge whether to link the mention *Black Panther* to the entity *Maykal* (The oldest feline) or the actor *Chadwick Boseman* whose nickname is *Black Panther*. Whereas, the image clearly indicates that the mention *Black Panther* is not an animal. It can be seen that without taking multimodal information into consideration, many valuable content would be lost and mentions cannot be correctly linked, especially when the text is relatively short and coarse. Hence, it is very valuable to incorporate multi-schema information when linking mentions to entities in KGs.

Although there have been extensive studies in entity linking [35], few work directly addresses the MEL, which is more challenging because of the mention ambiguity and multimodal information understanding. First, an entity may have various text mentions (e.g., *Chadwick Boseman* and *Black Panther*) and the associated images may contain multiple entities (e.g., actors and cities in the Marvel film poster), such ambiguities lead to difficulty in identifying correct entities. Second, understanding multimodal information requires the model to be able to utilize the intra-modality semantic information and mine the inter-modality associations. To solve the MEL task, a general approach is to use Multimodal Representation Learning (MRL), which exploits the complementarity and relationships of multiple modalities to represent and summarize multi-modal data. MRL has been proved useful in audio-video speech recognition, multimodal information retrieval, visual question answering (VQA), etc.

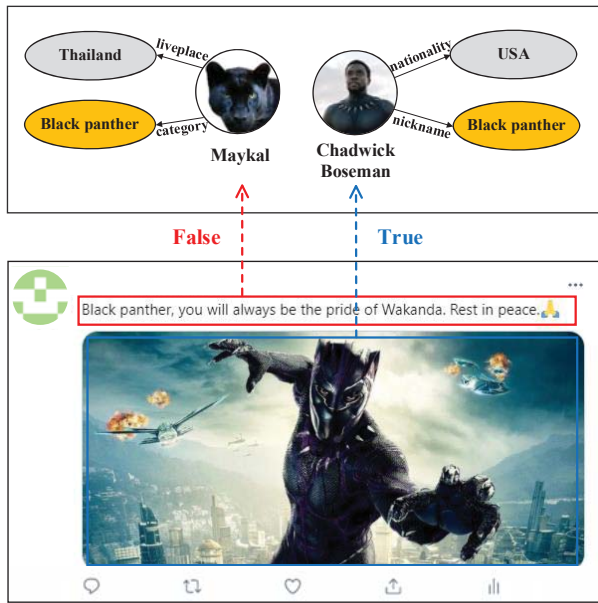


Figure 1: An example of multimodal entity linking in a tweet. *Black Panther* links to actor Chadwick Boseman.

Deep Boltzmann Machine (DBM) model for learning multimodal representations was proposed to benefit the image retrieval [37]. An auto-encoder model can improve the speech recognition by learning the unified representations of audio and video [30]. Since DBM and auto-encoder models have difficulties in extracting complementary information among different modalities, co-attention was used to discover supplementary and complementary relationships simultaneously between modalities [24]. In the multimodal named entity disambiguation task [28], a similar task to MEL, fusing the extracted features of text and images by attention mechanism could filter the noise contained in each modality. However, the co-attention mechanism in the above work tends to extract global rather than local interactions and cannot efficiently discover fine-grained relationships between modalities. Recent work on multimodal entity linking for tweets utilized fully connected layers to project the visual and textual features into an implicit joint space [2], but it ignored to model the inter-modal interactions and filter the noise contained in visual context features.

To tackle the above issues, this paper proposes a novel multimodal entity linking method GHMFC (Gated Hierarchical Multimodal Fusion and Contrastive training). GHMFC is intensively designed for learning the multimodal representations of mentions, which consists of two parts: hierarchical multimodal co-attention and gated fusion. First, to utilize multimodal information and discover fine-grained relationships across modalities, we introduce the feature extract structure of transformer [39] to implement the hierarchical multimodal co-attention: textual-guided visual attention and visual-guided textual attention. Textual-guided visual attention is used to highlight regions of visual information which are more related to text. On the contrary, visual-guided textual attention is used to highlight the positions of text which are more related

to visual information. Specifically, we extract visual features and two-level textual features (token-level and phrase-level) to calculate the hierarchical multimodal co-attention, respectively. Then, the weighted hierarchical features of text and vision are integrated into joint representations according to their importance to the task by gated fusion, which also mines the complementary information. In order to improve the quality of learned multimodal representation and reduce the noise, besides the triplet loss, we are among the first to adopt contrastive learning to MEL and design two types of losses: vision-text contrastive loss and text-vision contrastive loss. Finally, we link mentions to entities whose representations have the highest cosine similarity with the fused representations.

Moreover, to make up for the deficiency of open MEL datasets, this paper constructs two new open MEL datasets: WikiMEL and Richpedia-MEL. The WikiMEL has more than 22K multimodal samples extracted from Wikipedia and Wikidata [40]. The Richpedia-MEL is constructed based on the multimodal KG Richpedia [41] and contains more than 17K multimodal samples.

Experimental results show that using multimodal information leads to a significant improvement in MEL. Especially in Top 1-10 accuracy, GHMFC using visual information is 5-10% higher than the textual-only models. Further study demonstrates that GHMFC not only learns more valuable multimodal representations than the compared models, but also has better robustness to noises.

In summary, the main contributions of this paper are three-fold: (1) We focus on the rarely studied MEL task and propose a novel method GHMFC, which is capable of learning meaningful multimodal representations of mentions by gated fusion and contrastive training. (2) It introduces transformer to mine the fine-grained relationships across modalities in the MEL task, and utilizes contrastive learning to improve the quality of learned multimodal representation and reduce the noise. (3) Two new open MEL datasets WikiMEL and Richpedia-MEL are released and experimental results demonstrate that GHMFC outperforms previous state-of-the-art works.

2 RELATED WORK

Most previous EL methods mainly focus on exploiting the textual information from well-structured documents (e.g., AIDA CoNLL [20], IITB [21], N³ RSS500 [33], KORE [19], etc.), and can be divided into two groups: local methods [10, 25, 26] and global methods [16, 22, 32]. Local methods capture contextual information to disambiguate mentions. In contrast, global methods disambiguate mentions simultaneously by considering the topical coherence among a set of mentions. However, these methods cannot work well when processing multimodal data with not only text but also images and videos. Although there are few direct works on the MEL task, researchers have paid attention to multimodal representation learning (MRL), multimodal fusion, and multimodal attention mechanism, which are the related techniques of MEL.

Multimodal Representation Learning: To make use of multimodal information, MRL aims to automatically extract more comprehensive features by joint representations [7, 24, 37] and collaborative representations [13, 42]. Joint representation methods map multimodal data into the same feature space, in which each feature contains multiple modal information. A popular multimodal joint representation is the Deep Boltzmann Machines (DBMs) [37],

which is able to learn joint probability density from multimodal features and use hidden layer variables as input representation. However, DBMs has deficiencies in discovering complementary information among different modalities. In collaborative representation, each modal needs to train the single-modal representation separately, meanwhile, different modal representations need to satisfy some constraints [13]. However, collaborative representation still suffers from similarity measurements and structure constraints. Multimodal representation learning has been employed to quantify the entity coherence between image and text in real-world news [29]. Here, the named entity linking is applied to find persons, locations, and events in the news text. Reference data is automatically gathered from the Web and used in combination with novel measures of cross-modal similarity for the visual verification of entities in the article's photograph. From the MEL perspective, this work could be enlightening to improve the accuracy of MEL, and it is an unsupervised approach. In this paper, we borrow from both visions. Specifically, we first obtain the joint textual and visual representations as the semantic features of mentions, and then use the idea of collaborative representation to make the multimodal representations of mention close to the features of corresponding entity in KG.

Multimodal Fusion: Multimodal fusion considers the process of integrating information of different modalities as joint representations learning [3]. According to the levels of fusion, there are three multimodal fusion strategies: early fusion, late fusion and intermediate fusion. Early fusion directly concatenates the original features of each modality, which is simple but ignores the heterogeneity among modalities [14, 36]. By contrary, late fusion first learns unimodal features from different modalities and then fuses learned features into a multimodal semantic representation to avoid the problem of heterogeneous features fusion [14, 36]. The disadvantages of late fusion schemes are its expensiveness in terms of the learning effort and the potential loss of correlation in mixed feature space. Compared with the early fusion and late fusion strategies, intermediate fusion is a more flexible and general fusion strategy [11]. Specifically, each modal is transformed from a separate neural layer as an input and in some subsequent hidden layers, multiple modal representations are mapped into the joint space. Therefore, the intermediate fusion is able to capture interactions among modalities by learning intermediate features and adding fully connected layers to continue training the concatenated features [43].

Multimodal Attention Mechanism: By extending unimodal attention to multimodal attention to mine valuable information from other modalities, the attention mechanism has been widely used in many tasks [27]. In the multimodal named entity disambiguation task, Moon et al. [28] used modal attention mechanism for getting a multimodal representation of image, word and character. But it had deficiencies in discovering complementary information among different modalities. To address the above deficiencies, co-attention mechanism was used to calculate image attention and text attention vectors in order to get the weighted sum of visual features and textual features [24]. An adaptive co-attention mechanism was proposed to capture the textual and visual attention simultaneously [48]. Further, a dense co-attention mechanism is introduced for fine-grained interactions between text and image [45]. Zhang et al. [47]

consider bad effect caused by noisy images to MEL task, by identifying noisy images with a two-stage image and text correlation mechanism, and then design an attention mechanism to capture important information shared by the mention through querying multi-hop entities around candidate entities. There are two differences compared to our work: (1) Zhang et al.'s work focuses on the effect of image noise to MEL, while ours is a general model to solve MEL; (2) Zhang et al.'s attention mechanism is through querying multi-hop entities, while our GHMFC uses transformer to mine the fine-grained relationships across modalities in the MEL task. Despite the success of current methods are benefit for MEL task, they still fail to explore the fine-grained semantic interactions between different textual hierarchy and visual features. Moreover, the influence of noise caused by visual modal has not been intensively studied. To this end, this paper uses transformer to design a fine-grained multimodal co-attention mechanism and extracts two-level textual features to interact with visual information to discover hierarchical semantic interactions across multiple modalities.

3 METHODOLOGY

In this section, we first formally define the MEL. Then we give a brief overview of the proposed GHMFC model. Subsequently, we discuss the visual feature extraction and textual feature extraction. Furthermore, we elaborate the hierarchical multimodal co-attention and the gated hierarchical multimodal fusion. Finally, we address the entity linking calculation and the training of GHMFC in detail.

3.1 Preliminary

Let $X = \{x_i\}_{i=1}^N$ be a set of N input multimodal samples, with corresponding entities set $Y = \{y_i\}_{i=1}^M$ in a KG. Each input sample is composed of three parts: $x = \{x_w; x_v; x_m\}$, where $x_w = \{x_{w,t}\}_{t=1}^{L_w}$ denotes a text which consisting of a sequence of tokens with length L_w , x_v is the visual information associated with the text, and $x_m = \{x_{m,i}\}_{i=1}^{L_m}$ is a set of mentions with length L_m , respectively. The features extracted from textual and visual information are respectively denoted as f_t and f_v .

A KG is composed of a set of entities $\mathcal{E}$ and a set of relations $\mathcal{R}$. We define KG as $\mathcal{KG} = \{(h, r, t) | h, t \in \mathcal{E}, r \in \mathcal{R}\}$, where h is the head entity, t is the tail entity and each triple (h, r, t) indicates that there is a relationship r from h to t .

The MEL task aims to link mentions to entities in KGs and can be formulated as follows:

$$y = \arg \max_{y' \in Y} \text{sim}(\phi(x), \psi(y')) \quad (1)$$

where ϕ represents multimodal representation function, ψ represents the KG entities representation function and $\text{sim}(\cdot)$ produces a similarity score between the multimodal representation and ground truth KG entities.

3.2 Overview of GHMFC

The principle of GHMFC is shown in Figure 2. GHMFC includes three steps: (1) multimodal features extraction, (2) multimodal co-attention, and (3) gated hierarchical multimodal fusion. First, we respectively extract visual features by ResNet and embed token-level textual features with BERT. Following previous work [24],

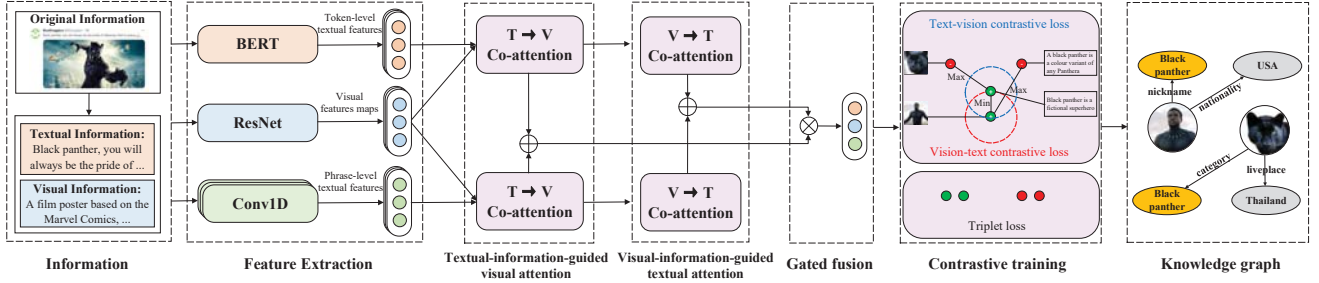


Figure 2: Overview of multimodal entity linking model based on GHMFC.

we convolve phrase embedding vectors using filters with different window sizes, and apply max-pooling across the various n-grams to obtain a single phrase-level representation. Afterwards, hierarchical multimodal co-attention mechanism alternately focuses on multimodal information, constructing visual-guided textual features and textual-guided visual features at two levels: token-level and phrase-level. We separately combine the guided two-level features of textual and visual information as their respective representations. Gated fusion is then used to adaptively integrate the above representations of textual modality and visual modality into a joint multimodal representation. We design vision-text contrastive loss and text-vision contrastive loss during contrastive training to improve the quality of multimodal representations and avoid noise in MEL. Finally, we obtain the linking entities by measuring the similarity between features of the multimodal representations and candidate entities, which are selected in advance by calculating the standardized edit distance between names of mentions and entities.

3.3 Candidate KG Entities Selection and Representation

In order to improve efficiency of the entity linking model, we first select candidate entities for each mention according to the edit distance between mentions and names of entities in KGs. For entities representations, we concatenate 14 main properties of an entity into a piece of text, which can be regarded as a summary of entity. For example, the summary of the wikidata [40] entity *Q3298076* is *Sex: male. Birth: 1962, Santa Rosa de cabal, ... , Occupation: football manager, football player*. Then the summary is input into a pre-trained BERT [9] model to encode the context information as the representation of the entity.

3.4 Multimodal Feature Extraction

3.4.1 Visual Feature Extraction. Visual features are extracted through the pretrained ResNet-200 [18]. In order to facilitate the fusion with textual features, a single-layer perceptron is used to map visual features into multimodal feature space:

$$f_v = \text{ReLU}(\tilde{f}_v W_v + b_v) \quad (2)$$

where $\tilde{f}_v$ is the output of ResNet-200, $f_v = \{V_i | V_i \in \mathbb{R}^{d_{model}}, i = 1, 2, \dots, |v|\}$ is the visual feature map after transformation by a single layer perceptron, d_{model} is the dimension of the multimodal feature space.

In order to capture the hierarchical semantic features, we extract two-level textual features.

3.4.2 Token-level Textual Feature Extraction. Token-level textual features are extracted with BERT and fine-tuned in training. We represent token-level textual features as: $\tilde{f}_w = \{\tilde{T}_i | \tilde{T}_i \in \mathbb{R}^{d_t}, i = 1, 2, \dots, |t|\}$, where $|t|$ represents the length of the sub-word pieces of text tokenized by BERT, $\tilde{T}_i$ represents the feature of i -th token, d_t is the dimension of $\tilde{T}_i$. A single-layer perceptron is used to map $\tilde{f}_w$ into the multimodal space:

$$f_w = \text{ReLU}(\tilde{f}_w W_w + b_w) \quad (3)$$

where W_w and b_w are trainable parameters.

3.4.3 Phrase-level Textual Feature Extraction. To capture the phrase-level textual features, the token-level embeddings obtained in the previous step are fed into various one-dimensional convolutions with different window sizes: unigram, bigram and trigram. Specifically, we set 3 filters $[C_1, C_2, C_3]$. The window sizes of these filters are $[1, 2, 3]$. Correspondingly, the phrase feature $\tilde{f}_p^k \in \mathbb{R}^{|t| \times d_{model}}$ generated with the k -th filter is obtained by the following operation:

$$\tilde{f}_p^k = [\dots, \text{ReLU}(C_k \cdot f_w, [i:i+k-1] + b_k), \dots] \quad (4)$$

where i is the index of the convolution window, f_w is the token-level embeddings of text, b_k is the bias term. To gain a more comprehensive representation of each token, max-pooling is applied across different n-grams at each token location, that finds the maximum value in the output of each filter. For i -th token, the corresponding phrase-level features is given by:

$$f_{p,i} = \max(\tilde{f}_{p,i}^1, \tilde{f}_{p,i}^2, \tilde{f}_{p,i}^3) \quad (5)$$

Then, phrase-level textual features are obtained by combining the phrase-level features of each token as follows:

$$f_p = [f_{p,1}, f_{p,2}, \dots, f_{p,|t|}] \quad (6)$$

3.5 Hierarchical Multimodal Co-attention

Multimodal co-attention (MCA) is designed to capture the fine-grained correlations between two modalities by calculating the attention weights between all textual positions and all visual regions. MCA is composed of two attention mechanism: textual-guided visual attention and visual-guided textual attention, which alternately guide each other to gather useful information from different modalities. Both of them are built upon the Cross-Modal Transformer (CMT) layer [38]. The most important component of CMT

is multi-head self-attention mechanism, which is stacked by scaled dot-product attention. Scaled dot-product attention takes query (Q), key (K), and value (V) as input, and then output the attended features as follows:

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

where $Q \in \mathbb{R}^{m \times d_k}$, $K \in \mathbb{R}^{n \times d_k}$, $V \in \mathbb{R}^{v \times d_k}$. Here, we set the dimension of Q , K and V to the same number d_k for simplicity. Given visual features Q and textual features K , $QK^T \in \mathbb{R}^{m \times n}$ can represent the dot-product similarity matrix of vision and text, and QK_{ij}^T represents the similarity between the j -th textual position and the i -th visual region, thus mining the fine-grained relationships between textual and visual information. Multi-head self-attention projects queries, keys and values to h different spaces and calculates scaled dot-product attention in parallel, and the output is obtained by concatenating the h results as follows:

$$\begin{aligned} \text{MH-SA}(Q, K, V) &= \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \\ \text{head}_i &= A(QW_i^Q, KW_i^K, VW_i^V) \end{aligned} \quad (8)$$

where $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_h}$, $W^O \in \mathbb{R}^{h \times d_h \times d_{\text{model}}}$ are projection matrices for the i -th head. $d_h = d_{\text{model}}/h$ is the dimension of each head.

Take *Red Bulls* in "Osorio during his time at New York Red Bulls" and a related image as an example, Figure 3 illustrates the attention weights at various regions of images by eight attention headers, in which higher weights mean more important regions. It can be seen that each attention header focuses on different regions. Header 1, 4, and 7 focus on the face and clothing. Header 3, 5, and 6 focus on the *Red Bulls* trademark on the background wall. Header 2 and 8 focus on the whole person. These eight attention headers are concatenated in subsequent calculations and fused via a linear transformation W^O to form the output of the layer.

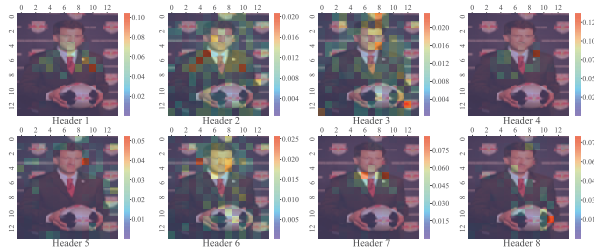


Figure 3: An example of multimodal co-attention with eight attention headers.

Besides, CMT also employs a feed-forward layer, residual connection and layer normalization to facilitate optimization. The whole operation of CMT can be described as follows:

$$\begin{aligned} \tilde{f} &= \text{LN}(Q + \text{MH-SA}(Q, K, V)), \\ f &= \text{LN}(\tilde{f} + \text{FFN}(\tilde{f})) \end{aligned} \quad (9)$$

where FFN is the feed-forward network, LN is the layer normalization, f is the output attended features. We first apply textual-guided visual attention to pay more attention to the visual regions which

are more related to the corresponding textual locations, obtaining the text-aware visual features as follows:

$$f_v^{\text{att}} = \text{CMT}(f_t, f_v, f_v) \quad (10)$$

Then visual-guided textual attention takes the text-aware visual features as the input queries, textual features as keys and values, obtaining vision-aware textual features as follows:

$$f_t^{\text{att}} = \text{CMT}(f_v^{\text{att}}, f_t, f_t) \quad (11)$$

Furthermore, to mine the fine-grained correlations from different semantic hierarchy, we extend the MCA mechanism to token-level and phrase-level MCA. Token-level MCA gets the input textual features f_w obtained by Equation (3) and visual features f_v obtained by Equation (2), producing token-level attended visual features $f_{v:w}^{\text{att}}$ and textual features $f_{t:w}^{\text{att}}$. Similarly, phrase-level MCA outputs phrase-level attended visual features $f_{v:p}^{\text{att}}$ and textual features $f_{t:p}^{\text{att}}$. Afterwards, we concatenate the hierarchical features of each modality as follows:

$$\hat{f}_t = \text{Concat}(f_{t:w}^{\text{att}}, f_{t:p}^{\text{att}}) \quad (12)$$

Then max-pooling is again used to discover the most important features of each dimension:

$$f_t^i = \max(\hat{f}_{t,1}^i, \hat{f}_{t,2}^i, \dots, \hat{f}_{t,2|t|}^i) \quad (13)$$

Then we combine features of each dimension to obtain final textual features as follows:

$$f_t = [f_t^1, f_t^2, \dots, f_t^{d_{\text{model}}}] \quad (14)$$

Following similar steps, the combined visual features $f_v \in \mathbb{R}^{d_{\text{model}}}$ can be obtained.

3.6 Gated Hierarchical Multimodal Fusion

Based on the hierarchical multimodal co-attention, GHMFC adaptively assesses the contributions of features in each modality to the MEL task, and fuses them together in different proportions. Concretely, GHMFC utilizes gated fusion to learn two linear mappings to map f_t and f_v to h_{f_t} and h_{f_v} , with ReLU as the activation functions:

$$h_{f_t} = \text{ReLU}(f_t \cdot W_{f_t} + b_{f_t}) \quad (15)$$

$$h_{f_v} = \text{ReLU}(f_v \cdot W_{f_v} + b_{f_v}) \quad (16)$$

where $W_{f_t}, W_{f_v} \in \mathbb{R}^{d_{\text{model}}}$, b_{f_t} and b_{f_v} are learnable parameters. Then, it concatenates the results of the previous step and uses another linear mapping and softmax to map the concatenated features to $g \in \mathbb{R}^2$ as follows:

$$g = \text{softmax}([h_{f_t}, h_{f_v}] \cdot W_g + b_g) \quad (17)$$

where W_g and b_g are learnable parameters. The first dimension g_1 of g represents the importance of textual features, and the second dimension g_2 represents the importance of visual features, where $g_1 + g_2 = 1$. Finally, the multimodal representation f_m is obtained by calculating the weighted sum of f_t and f_v with weights g :

$$f_m = g \cdot [h_{f_t}, h_{f_v}]^T \quad (18)$$

After learning multimodal representations of mentions, we use cosine similarity to measure the similarity between the mention and the candidate entities. The entity with the highest similarity to the mention is regarded as the corresponding linking entity.

3.7 Contrastive Training

We use contrastive learning idea in training to improve the quality of learned multimodal representations and reduce the noise. Our training scheme is based on self-supervised representation learning framework MoCo-v2 [5]. We design two types of contrastive losses: (1) vision-text contrastive loss and (2) text-vision contrastive loss. Given an vision-text pair (v_i, t_i) , we aim to simultaneously minimize the distance between $f_m(v_i)$ and $f_m(t_i)$ and maximize the distances between $f_m(v_i)$ and all other negative features. For example in Figure 2, we aim to fine tune related vectors of the image and the text about Chadwick Boseman to be close, while staying away from the unrelated vectors. The vision-text contrastive loss is defined as:

$$\mathcal{L}_{vt} = \sum_{j=1}^K (f_m(v_i) \cdot f_m(t_j)^+ - f_m(v_i) \cdot f_m(t_j)^-) \quad (19)$$

where + denotes the positive sample, - denote the negative samples. Similarly, the text-vision contrastive loss is:

$$\mathcal{L}_{tv} = \sum_{j=1}^K (f_m(t_i) \cdot f_m(v_j)^+ - f_m(t_i) \cdot f_m(v_j)^-) \quad (20)$$

To this end, the final loss for our multimodal contrastive training model is:

$$\mathcal{L}_c = \lambda_{vt} \mathcal{L}_{vt} + \lambda_{tv} \mathcal{L}_{tv} \quad (21)$$

where λ_{vt} and λ_{tv} are trade-off parameters in two contrastive losses.

We not only try to increase the similarity between the mention and its corresponding entity (positive sample) in KGs, but also lower the similarity between the mention and other entities (negative sample). The contrastive losses and generic triplet loss are used as an optimization goal for the MEL task. The optimization goal is defined as follows:

$$\begin{aligned} \min (\mathcal{L}_c + \mathcal{L}(X, Y^+, Y^-; W) + \frac{\alpha}{2} \|W\|_2), \\ \mathcal{L}(X, Y^+, Y^-; W) = \frac{1}{N} \sum_{i=1}^N \max\{0, S(F(x_i; W), G(y_i^-)) \\ - S(F(x_i; W), G(y_i^+)) + m \end{aligned} \quad (22)$$

where $\mathcal{L}(\cdot)$ is the triplet loss, X represents the set of multimodal data, Y^+ and Y^- represent the positive and negative entity sets, W is the learnable parameters, $S(\cdot)$ is the cosine similarity function, $F(\cdot)$ and $G(\cdot)$ represent the multimodal feature extraction and entity feature extraction, m is the safety interval between the similarity of positive samples and negative samples. $\frac{\alpha}{2} \|W\|_2$ is a regularization term and α is the coefficient of the regularization term.

3.8 Entity Linking Calculation

After learning multimodal representations of mentions, we use cosine similarity to measure the similarity between the mention and the candidate entities. The entity with the highest similarity to the mention is regarded as the corresponding linking entity. For example, in a multimodal sample, the text information is *Williams performing stand-up comedy at a USO show on December 20, 2007*, the visual information is an image of a man holding a microphone, and the mention is *Williams*, and the candidate entities are *Robin Williams (stand-up comedian. 1951)*, *Billy Dee Williams (film actor, novelist. 1937)*, *Olivia William (film actor, playwright. 1968)*, *John*

Williams (pianist, composer. 1932), and *William M. Robbins (lawyer, politician. 1905)*. By calculating the cosine similarity, the mentioned similarities with the entities were obtained as 0.88, 0.77, 0.51, 0.41 and 0.33, respectively. Therefore, *Robin Williams (stand-up comedian. 1951)* is chosen as the linked entity with the highest similarity. We obtain candidate entities in three steps: (1) Exact Matching: Using the longest common subsequence algorithm to mark the public part of text information and entity names; (2) Fuzzy Matching: Analyzing the common prefix and normalized edit distance between text information and entity names, and strings with common prefix length greater than zero and normalized edit distance less than a threshold are marked as mentions; (3) Abbreviation Matching: Finding abbreviations of entity names and mark them as mentions. After obtain a candidate entity, we first form a summary with all the neighbor nodes of the entity in the KG. For example, the abstract of a Wikidata entity *Q3298076* is *Sex: male: Football manager, football player....* Then, we use pre-trained BERT model to extract the sentence features from the summary. Such the sentence feature is represented as a vector. Therefore, it can be used to calculate the cosine similarity.

4 EXPERIMENTS

This section reports the experimental results. We first address the experimental settings, including the datasets, baseline methods, parameter settings, and evaluation metrics. Then, we compare GHMFC with state-of-the-art MEL solutions on the datasets. Furthermore, we conduct the ablation study and detailed analysis of the components in GHMFC. Subsequently, we analyze the robustness of MEL models and study the performance of MEL models under different noises. Finally, we perform detailed case study and limitation analysis. The code of GHMFC and the datasets can be accessed via <https://github.com/seukgcode/MEL-GHMFC>.

4.1 Datasets, baselines and settings

4.1.1 Datasets. To the best of our knowledge, MEL task still lacks accessible datasets. A dataset for multimodal named entity disambiguation [28] could be used in MEL, but it is not public. Recently, we notice that a new dataset has emerged, but until now it has not been publicly available [15]. Adjali et al. proposed a method for building a MEL dataset from Tweets [1] and then evaluated their MEL model [2]. However, due to the expiration of a large amount of Twitter content, this MEL data set cannot be fully reproduced. Fortunately, according to the provided source codes¹ and building process [2], we create a Twitter-MEL-like dataset with more than 31K multimodal samples from Tweets. Furthermore, we release two new MEL datasets: WikiMEL and Richpedia-MEL. For WikiMEL, we first collect entities from Wikidata [40], and then extract the textual and visual descriptions of each entity from Wikipedia. WikiMEL has more than 22K multimodal samples. For Richpedia-MEL, we first collect the Wikidata ids of entities in a large-scale multimodal knowledge graph Richpedia [41], then obtain the corresponding multimodal information from Wikipedia pages. Richpedia-MEL has more than 17K multimodal samples. The datasets contain common entity types: person, location, organization, etc. The KG used in this paper is a subset extracted from Wikidata, and has more than 170K

¹https://github.com/OA256864/MEL_Tweets

triples and about 80K entities, which contains all entities in datasets. To control the quality of new datasets, based on the candidate mentions obtained by the annotation algorithm, mentions are further verified by five annotators. Four annotators verify and correct raw mentions independently, and the fifth randomly check the verified mentions. Finally, low confident annotations are discussed by all annotators.

The statistics of datasets are summarized in Table 1, which contains total samples, mentions, average text length of a sample, and average number of mentions in a sample. We further analyze the text length and mention number distributions of datasets as Figure 4 shown. According to Figure 4(a), it can be seen that most texts in WikiMEL and Richpedia-MEL are composed of 2-12 words, and most Twitter-MEL texts have 10-20 words. It means that, besides the short textual information, multimodal information could be useful for MEL task. Figure 4(b) depicts the number of mentions distribution in a text. It can be seen that most samples contain a single mention and few samples contain multiple mentions, meanwhile, there are some samples that do not contain any mentions. In our experiments, these datasets are randomly divided into training (70%), validation (10%), and testing sets (20%).

Dataset	Samples	Mentions	Text length (avg.)	Mentions (avg.)
WikiMEL	22,136	25,846	8.2	1.2
Richpedia-MEL	17,806	18,752	13.6	1.8
Twitter-MEL	31,304	33,394	15.6	1.5

Table 1: Summary statistics for the datasets.

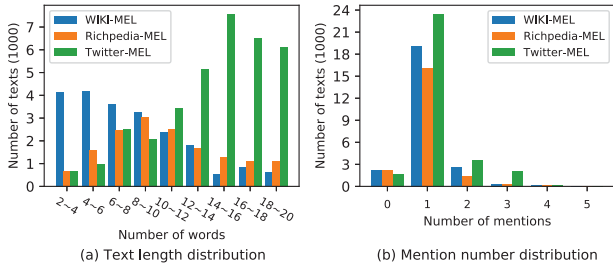


Figure 4: Text length and mention distribution of datasets.

4.1.2 Baselines. We compare GHMFC with some strong baselines, which are divided into two groups: (1) text only approaches that use textual features; and (2) text + vision approaches that use textual and visual features.

In text only group, we select ARNN [12], BERT [9], BLINK [44], and GENRE [8] as baselines to evaluate the best performance that can be achieved with only textual modality.

- **ARNN** (text only) [12] utilizes the *Attention-RNN* to predict associations with candidate entity textual features. We note that there is usually noise in the text of the MEL. ARNN is a named entity disambiguation method for noisy text. In our paper and other previous works such as DZMNED [28], ARNN is used as a baseline for EL in noise text.

- **BERT** (text only) [9] stacks several layers of transformer to encode each token in the text. In this paper, BERT is used as a

text encoder to capture textual features. In particular, to ensure the fairness in the evaluation and to verify that the visual modality can improve MEL performance, the text + vision approaches also employ the textual features obtained by BERT (such as DZMNED-BERT).

- **BLINK** (text only) [44] is an BERT-based EL model with a two-stage zero-shot linking.

- **GENRE** (text only) [8] is an autoregressive end-to-end EL model via a sequence-to-sequence architecture to generate entity names in an autoregressive fashion.

In the text + vision group, we select HieCoATT-Alter [24], DZMNED [28], JMEL [2], and MEL-HI [47] as comparison baselines.

- **HieCoATT-Alter** (text + vision) [24] is a multimodal fusion mechanism, and it uses *alternating co-attention* and three textual levels (tokens, phrases, and sentences) to calculate co-attention maps. Multimodal feature fusion is critical for the MEL task. In this paper, we use this mechanism in the MEL and then implement the baseline HieCoATT-Alter.

- **DZMNED** (text + vision + character) [28] is among of the first work to address the MEL task, and utilizes a concatenated multimodal attention mechanism to fuse visual, textual and character features of mentions and entities.

- **DZMNED-BERT** (text + vision + character) For fairness, we also replace the GLOVE pre-trained word vector of GLOVE [31] originally used by DZMNED with BERT.

- **JMEL** (text+vision) utilizes fully connected layers to project the visual and textual features into an implicit joint space [2].

- **MEL-HI** (text + vision) [47] uses multiple attention mechanisms to get richer information and removes the negative impact of noisy images.

- **GHMFC-OnlyText** (text only) The variant of our model with only textual features.

- **GHMFC-Alter** (text + vision) The variant of our model by replacing the multimodal co-attention mechanism with the *alternating co-attention* [24].

- **GHMFC-GatedText** (text + vision) The gated fusion module is removed, and the textual feature after multimodal fusion is directly used as the semantic features of mentions.

- **GHMFC** (text + vision) This is the proposed MEL model.

4.1.3 Metrics. We use Top-k accuracy as the metric that can be calculated by the following formula:

$$\text{Accuracy}_{top-k} = \frac{1}{N} \sum_{i=1}^N I(t_i \in y_i^k) \quad (23)$$

where N represents the total number of samples, and I is the indicator function. When the receiving condition is satisfied, I is set to 1, and 0 otherwise.

4.1.4 Hyperparameters and Training Details. We set the dimensions of textual and visual features to 512, MCA dimension to 512, the number of stacked layers of multimodal co-attention to 2, mention feature and entity feature dimensions to 768, heads of MH-SA to 8, dropout to 0.4, triplet loss interval to 0.5. We optimize the parameters with AdamW [23] with batch size 32, learning rate 5×10^{-5} , L2 regularization coefficient 0.2 and gradient clipping threshold 1.0.

Modalities	Models	WikiMEL				Richpedia-MEL				Twitter-MEL			
		Top-1	Top-5	Top-10	Top-20	Top-1	Top-5	Top-10	Top-20	Top-1	Top-5	Top-10	Top-20
T	ARNN [†] [12]	32.0	45.8	56.6	65.0	31.2	39.3	45.9	54.5	51.8	59.1	64.6	72.7
T	BERT [†] [9]	31.7	48.8	57.8	70.3	31.6	42.0	47.6	57.3	54.3	60.4	65.9	75.0
T	BLINK [†] [44]	30.8	44.6	56.7	66.4	30.8	38.8	44.5	53.6	50.6	58.7	64.1	71.5
T	GENRE [†] [8]	32.5	49.2	58.5	71.8	32.0	42.3	48.5	59.3	54.7	61.5	66.2	76.5
T + V	JMEL [2]	31.3	49.4	57.9	64.8	29.6	42.3	46.6	54.1	55.1	62.9	66.3	73.8
C + T + V	DZMNED [28]	30.9	50.7	56.9	65.1	29.5	41.6	45.8	55.2	54.0	62.5	66.6	76.7
C + T + V	DZMNED-BERT	34.7	53.9	58.1	70.1	32.4	43.7	48.2	60.8	54.7	63.4	68.7	78.1
T + V	HieCoATT-Alter [24]	40.5	57.6	69.6	78.6	37.2	46.8	54.2	62.4	55.7	68.7	71.3	80.2
T + V	MEL-HI [47]	38.6	55.1	65.2	75.7	34.9	43.1	50.6	58.4	50.9	63.5	66.7	74.8
T	GHMFC-OnlyText	34.1	51.3	60.4	72.5	31.8	43.3	48.8	59.3	53.4	63.7	69.8	79.9
T + V	GHMFC-Alter	42.9	61.2	71.4	81.7	39.0	49.5	55.6	64.0	60.5	68.8	73.4	82.4
T + V	GHMFC-GatedText	40.6	64.2	73.1	87.1	38.4	49.8	56.4	65.2	59.9	66.5	70.1	80.1
T + V	GHMFC	43.6	64.0	74.4	85.8	38.7	50.9	58.5	66.7	61.3	70.3	76.2	84.5

Table 2: MEL results of the compared models at Top-1, 5, 10, 20 accuracies (%). (T: textual modal, V: visual modal, C: character). [†] stands for results produced by original implementations. Other baselines are implemented according to corresponding papers.

4.2 Main Experimental Results

We conduct comparative experiments to evaluate the effectiveness of each model in the MEL task. Table 2 shows the Top-1, 5, 10, 20 accuracy results of all models on three datasets.

First, we observe that adding visual features can improve the performance of MEL. Models that incorporate visual information such as HieCoATT-Alter, GHMFC, JMEL, MEL-HI, etc., generally outperform textual-only models such as ARNN, BERT, BLINK, GENRE, and GHMFC-OnlyText on all datasets. Compared with GHMFC-OnlyText, GHMFC has 4–13% improvement in each metric. In particular, the Top-10 accuracy of the GHMFC is 13.8%, 9.7%, and 6.4% higher than the GHMFC-OnlyText on WikiMEL, Richpedia-MEL, and Twitter-MEL, respectively. Therefore, it can be seen that adding visual features can improve the accuracy of MEL.

Second, it is noticed that multimodal representation learning method can improve the performance of MEL. As shown in Table 2, the GHMFC models group as well as the HieCoATT-Alter outperform the DZMNED models group and JMEL, indicating that using multimodal co-attention mechanism can effectively mine the relationships between multiple modalities. DZMNED-BERT uses a simple concatenation approach to discover correlations between multimodal features, which provides limited improvement over the co-attention used by models such as HieCoATT-Alter and GHMFC. Top-1 accuracy of DZMNED is even lower than BERT, suggesting that simply concatenating heterogeneous multimodal data may even make the model worse. JMEL has similar performance as BERT, and it reveals that fully connected layers have limited ability to capture modals association information. Furthermore, GHMFC outperforms GHMFC-Alter on nearly all metrics. It suggests that the multimodal co-attention mechanism used by GHMFC is able to exploit the associations between textual locations and visual regions at a finer granularity, bringing in richer semantic information.

Third, GHMFC achieves the state-of-the-art performance in most metrics on all datasets. The baseline models are compared from the aspects of the function of each modal and the extraction method of modals association information. Specifically, GHMFC has the highest accuracy in Top-1 and Top-10 on WikiMEL, Top-5 to Top-20 on Richpedia-MEL, and all metrics on Twitter-MEL. Compared with ARNN, BERT, BLINK, GENRE and MEL-OnlyText, which

only use textual information, GHMFC has nearly 10% improvement in most metrics. Compared with JMEL, MEL-HI, DZMNED, HieCoATT-Alter and GHMFC-Alter, which use multimodal information, GHMFC also has 1-8% improvement.

Finally, the experimental results demonstrate that textual features are effective in simple MEL tasks, but not effective in difficult MEL tasks. Here, Top-1 is the most difficult MEL task, and Top-20 is the simplest MEL task. Obviously, the more difficult the problem is to solve, the more fine-grained the model can distinguish. It can be found that the performance gap between models using textual features and the models using multimodal features is gradually decreasing from difficult tasks to simple tasks. Comparing the performance of DZMNED-BERT and BERT, DZMNED-BERT has a 3-4% improvement in the accuracy of Top-1 and Top-5, but BERT is even higher than the former in the accuracy of Top-20. Comparing the performance of models with similar structure: GHMFC-OnlyText and GHMFC, the difference between them is gradually narrowing in the Top-10 and Top-20 accuracy. Comparing GHMFC-GatedText and GHMFC, the former is superior to the standard model in the Top-20 accuracy. It means that the textual information only plays an important role in Top-10 and Top-20. The reason may be that the textual information can only distinguish positive and negative samples in a coarse-grained way, but it cannot work well in a fine-grained way with only limited textual information. In contrast, we also notice that adding visual information could make models have the ability of fine-grained distinguishing in difficult MEL tasks.

4.3 Ablation Experiments

We conduct ablation experiments on WikiMEL to demonstrate the validity of each module of GHMFC. Table 3 shows the performance of nine ablation models.

Compared with model 1 which only uses text, the Top-1 accuracy of GHMFC (model 10) is improved from 36.3% to 43.6%, and other metrics are also optimized by nearly 6%. The models using visual information (except model 3) are also improved by at least 3% on all metrics. It illustrates that visual information can provide additional semantic information, and models using multimodal information can yield better results than the model using only textual features.

The MCA mechanism proposed in this paper plays a key role in MEL, and GHMFC improves by 1-3% on each metric compared

No.	M	CA	WH	GF	CT	Accuracy (%)					
						Top-1	Top-3	Top-5	Top-10	Top-20	Top-50
1	T	Y*	To + P	Y	Y	36.3	50.6	57.5	68.9	78.7	90.0
2	T + V	Y*	To + P + S	Y	Y	42.0	53.8	63.4	72.5	83.2	92.1
3	T + V	N	To + P	Y	Y	37.3	49.3	57.7	68.9	78.4	89.4
4	T + V	Y	To	Y	Y	41.4	53.7	62.2	73.3	85.4	92.7
5	T + V	Y	P	Y	Y	41.7	54.4	62.5	73.6	83.5	90.9
6	T + V	Y	To + P + S	Y	Y	40.9	52.9	61.4	72.3	83.0	90.1
7	T + V	Y	To	N	Y	41.4	54.2	62.5	73.3	82.8	91.6
8	T + V	Y	To + P	Y*	Y	41.6	54.7	62.7	74.4	85.2	94.0
9	T + V	Y	To + P	Y	N	41.5	54.5	61.8	73.2	82.5	92.6
10	T + V	Y	To + P	Y	Y	43.6	56.6	64.0	74.4	85.8	95.9

Table 3: Results of GHMFC ablation experiments on WikiMEL (M: Modal used, CA: Co-Attention, HT: Hierarchy of Text, GF: Gated Fusion, CT: Contrastive Training, T: Textual modal, V: Visual modal, To: Token-level, P: Phrase-level, S: Sentence-level, Y: Yes, N: No, * means that the structure of the module is modified).

with model 2, which utilizes an alternating co-attention mechanism. It suggests that MCA is capable of generating more meaningful multimodal representations.

In previous methods, each modality is first mapped as a vector of R^d and then mined correlations in vector space, which could lose much detailed information. In contrast, GHMFC preserves the feature map structure of each modality and uses the transformer to mine the associations between modalities at a fine-grained level, namely, the hierarchical multimodal co-attention.

Hierarchical MCA also gives GHMFC a boost in performance. As shown in Table 3, model 4 (token-level only) and model 5 (phrase-only) have similar performance. Utilizing both token-level and phrase-level textual features, GHMFC has more than 1.5% improvements compared with model 5 on all metrics, especially on Top-1, Top-3, and Top-5 accuracies, but has similar performance on Top-20 and Top-50 accuracies. This shows that the combining multimodal features can effectively distinguish difficult samples.

In addition, we also find that gated fusion mechanism exploits complementary information between different modalities. Compared with GHMFC, the performance of Model 7 on all metrics is at least 2% lower than that of GHMFC, while Model 8 has worse performance than GHMFC at Top-1 to Top-5 accuracies, but exceeds GHMFC at Top-10 and Top-20 accuracies. Therefore, the gated hierarchical multimodal fusion automatically learns the importance of a modality and dynamically controls the impact of each modality on the final results.

It is noticed that the performance of model 6 with sentence-level features is at least 2.7% lower than other models. It is found that the results of QK^T in Equation 8 are scalars when calculating the image attention guided by textual information. The reason is that visual information is only relevant to a few words in a sentence, thus incorporating sentence-level features may bring additional noise to the subsequent fusion, which causes a lower accuracy when multimodal features are used simultaneously. Therefore, GHMFC exploits both token-level and phrase-level features, while abandoning sentence-level features. In particular, the results of model 9 show that without contrastive training, the performance decreases significantly. This suggests that contrast learning does improve the performance of the MEL task.

According to the ablation experimental results, in our GHMFC model, the key to improving performance lies in two aspects: (1)

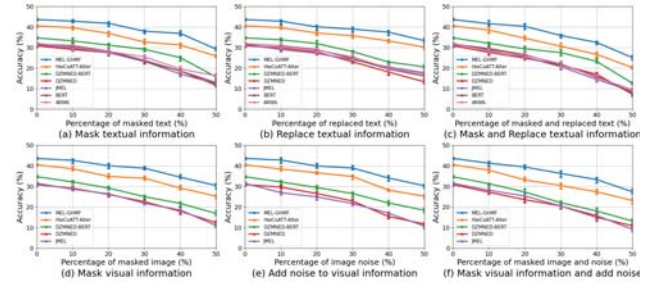


Figure 5: Results of robustness analysis experiments on WikiMEL (All results are average Top-1 accuracies).

Mining the correlations between modalities by the multimodal co-attention (MCA) mechanism, and (2) Using the gated hierarchical multimodal fusion to adaptively assess the contributions of features in each modality and fuse them together in different proportions. The gated fusion mechanism is able to mine complementary information from each modality, and the multimodal features can effectively distinguish difficult samples.

4.4 Robustness Analysis Experiments

To explore the robustness of MEL models, we study the performance of MEL models under different noises. We first keep the visual information unchanged, and then use three operations to randomly add noise to text: (1) mask part of the text, (2) replace some words, and (3) mask text and replace words at the same time. Afterwards, we keep the textual information unchanged and also use three similar operations to add noise to vision: (1) mask part of the vision, (2) inject Gaussian noise to vision, and (3) mask vision and inject noise at the same time.

As shown in Figure 5(a), 5(b) and 5(c), when text noise increases, performances of models (e.g., ARNN, BERT, BLINK and GENRE) without visual information decrease more quickly than models with visual information. It indicates that visual information can compensate for the lack of textual information. According to Figure 5(b) and 5(c), in which the texts are randomly replaced or masked from 10% to 40%, GHMFC has the least loss in accuracy compared with other models. That suggests that GHMFC has better robustness to text noise.

Case		1	2	3	4	5	6	7
Multimodal Data	Visual Information							
	Textual Information	Steve Spurrier was recognized as the game's Most Valuable Player	Time named Sivan as one of the "25 Most Influential Teens of 2014"	Williams performing stand-up comedy at a USO show on December 20, 2007	Robinho playing for Brazil vs Chile at the 2010 World Cup	Jung at the press conference for drama Cheer Up! in October 2015	Moravec pictured in ministerial uniform wearing the armband of the Kuratoriu in 1965	Russian President Vladimir Putin with Gabriel, 9 March 2017
	Entity	Steve Spurrier	Sivan	Williams	Robinho	Jung	Moravec	Vladimir Putin
MEL Prediction (Top-k)		1. Steve Spurrier 11.6 (American football player. 1945) 2. Thomas Hitzsperger 10.4 (association football player. 1982) 3. Daniel Sturridge 10.2 (association football player. 1989) 4. Carrie Brownstein 9.9 (actor, singer. 1974) 5. Jane Preiser 2.1 (actor. 1920)	1. Troye Sivan 12.7 (singer, youtuber. 1995) 2. Maureen O'Sullivan 8.8 (film actor, stage actor. 1911) 3. Khloé Kardashian 6.4 (actor, author, model. 1984) 4. Ivan Mojskovic 5.0 (actor, stage actor. 1889) 5. Sebastian Coe 4.9 (politician, runner. 1956)	1. Robin Williams 8.8 (stand-up comedian. 1951) 2. Billy Dee Williams 7.7 (film actor, novelist. 1937) 3. Olivia Williams 5.1 (film actor, playwright. 1968) 4. John Williams 4.1 (pianist, composer. 1932) 5. William M. Robbins 3.3 (lawyer, politician. 1905)	1. Ronaldinho 14.7 (association football player. 1980) 2. Robinho 13.6 (association football player. 1984) 3. Erwin Hoffer 12.9 (association football player. 1987) 4. Thomas Breich 12.8 (association football player. 1981) 5. Robin Dutt 12.8 (association football player. 1965)	1. Choi Soo-young 14.9 (actor, model, singer. 1990) 2. Lee Min-jung 14.3 (actor, film actor. 1982) 3. Daesung 14.0 (actor, singer. 1989) 36. Jung Eun-ji ... (actor, dancer, singer. 1993) 1.2	1. Giacomo Durando 9.8 (military personnel. 1807) 2. Alberto Moravia 9.1 (politician, journalist. 1990) 3. Nikolay Muraviev 9.0 (diplomat, politician. 1850) ... 8. Emanuel Moravec 8.6 (politician, writer. 1893)	1. Lyudmila Putina 15.4 (teacher, politician. 1916) 2. Valery Rysmin 14.7 (astronaut, engineer. 1939) 3. Sandro Pertini 13.7 (journalist, politician. 1896) ... 39. Vladimir Putin 5.9 (politician, economist, jurist. 1968)

Figure 6: Multimodal entity linking cases (Each case contains visual information, textual information, and an entity. The linkable mention is in bold blue text. The MEL results are ranked and the correct answer is in bold blue text).

From Figure 5(d), 5(e) and 5(f), it can be seen that the accuracy of the GHMFC model decreases from 43.6% to nearly 30% with vision noise increasing, but it is smaller than that of other models. It means that GHMFC has higher stability in processing visual information compared to other models. Figure 5(e) clearly reveals that the Top-1 accuracy of JMEL declines from 31.3% to 10%, which is greater than the models using multimodal attention mechanism, illustrating the effectiveness of multimodal attention mechanism in filtering the noise caused by visual information. In Figure 5(e) and Figure 5(f), all models get a sharp decrease in 40% and 50% vision noise. The reason is that too much noise could cause models to learn wrong correlations between textual and visual information.

4.5 Case Study

We further analyze the effectiveness of GHMFC by cases in Figure 6, which are divided into three groups: correct prediction, wrong prediction and mismatch.

Case 1, Case 2, and Case 3 are correct predictions. In Case 1 and Case 2, the text contains mentions or time information, while the image has features that can benefit disambiguation. For Case 2, GHMFC prioritizes entities whose professions are actors, singers, and politicians based on the *microphone* in the image, and interestingly, it further gives the correct linking results according to the time information in text. In Case 3, GHMFC mines the relationship between text and image: *microphone* and the *performing*, as well as the relationship in text: *comedy* and *show*. And it correctly identifies the entity as a stand-up comedian for the reason that microphone, performing and show often corresponds, rather than the singer, actor, etc. These correct prediction cases show that GHMFC is capable of disambiguating only by the visual information (Case 1), combining the textual and visual information (Case 2), and mining the relationships between multiple modalities (Case 3).

Case 4 and Case 5 are wrong predictions. In Case 4, GHMFC fails to differentiate between the entity *Robinho* and *Ronaldinho* (both are Brazilian soccer players). In Case 5, the visual information is used by GHMFC to rank the entities whose occupations are singers and actors, but it is not able to distinguish them due to the large number of similar entities in candidate entities.

The multimodal information and KGs could mismatch. For Case 6, although *Moravec* is known to be a soldier by associating textual and visual information, the entity *Emanuel Moravec* in a KG does not have a soldier's occupation, which causes a linking error. For Case 7, *Russian President* in the text directly indicates the identity of the mention *Vladimir Putin*, but this information is not provided in the KG. In addition, since the named entity recognition does not recognize *Gabriel*, GHMFC fails to link the mention to any entity.

5 CONCLUSIONS AND FUTURE WORK

We propose a multimodal entity linking method GHMFC, which is capable of learning relevant and complementary information across modalities at a fine-grained level. GHMFC can utilize the textual and visual information to disambiguate the mentions in a fine-grained way, mine the hidden and explicit relationships across modals, and reduce the noise of each modal via contrastive training. Experimental results show that GHMFC outperforms the state-of-the-art models. Specially, two new open MEL datasets WikiMEL and Richpedia-MEL are released to alleviate the lack of accessible datasets in MEL task.

Since the GHMFC is a pipeline model, a limitation of the method is that the error propagation in each module could affect the final MEL performance. Moreover, like other MEL models, GHMFC is difficult to link mentions to entities in KGs without sufficient textual and visual information. In addition, GHMFC need a KG to cover the entities in multimodal data to avoid the mismatch problem. In the future work, we plan to extend GHMFC to an end-to-end model to avoid the error propagation problem. Besides, even this paper mainly discusses textual and visual modalities, our method is able to be extended to process more modalities.

ACKNOWLEDGMENTS

We thank the anonymous reviewers for their insightful comments. The work was supported in part by the 13th Five-Year All-Army Common Information System Equipment Pre-Research Project (Grant Nos. 31514020501, 31514020503). All opinions are of the authors and do not reflect the view of sponsors.

REFERENCES

- [1] Omar Adjali, Romaric Besançon, Olivier Ferret, et al. 2020. Building a Multimodal Entity Linking Dataset From Tweets. In *Proceedings of the 12th Conference on Language Resources and Evaluation*. 4285–4292.
- [2] Omar Adjali, Romaric Besançon, Olivier Ferret, et al. 2020. Multimodal Entity Linking for Tweets. In *Proceedings of the 42nd European Conference on Information Retrieval*. 463–478.
- [3] Pradeep K Atrey, M Anwar Hossain, Abdulmotaleb El Saddik, et al. 2010. Multimodal Fusion for Multimedia Analysis: A Survey. *Multimedia systems* 16 (2010), 345–379.
- [4] Roi Blanco, Giuseppe Ottaviano, and Edgar Meij. 2015. Fast and space-efficient entity linking for queries. In *Proceedings of the 8th ACM International Conference on Web Search and Data Mining*. 179–188.
- [5] Xinlei Chen, Haoqi Fan, Ross B. Girshick, and Kaiming He. 2020. Improved baselines with momentum contrastive learning. *arXiv preprint* (2020), 2003.04297.
- [6] Andrew Chisholm and Ben Hachey. 2015. Entity Disambiguation with Web Links. *Transactions of the Association for Computational Linguistics* 3 (2015), 145–156.
- [7] Guillem Collell, Ted Zhang, and Marie-Francine Moens. 2017. Imagined Visual Representations as Multimodal Embeddings. In *Proceedings of the 31th AAAI Conference on Artificial Intelligence*. 4378–4384.
- [8] Nicola De Cao, Gautier Izacard, Sebastian Riedel, and Fabio Petroni. 2021. Autoregressive Entity Retrieval. In *International Conference on Learning Representations (ICLR)*.
- [9] Jacob Devlin, Ming-Wei Chang, Kenton Lee, et al. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 17th Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 4171–4186.
- [10] Stephen Dill, Nadav Eiron, David Gibson, et al. 2003. SemTag and Seeker: Bootstrapping the Semantic Web via Automated Semantic Annotation. In *Proceedings of the 12th International Conference on World Wide Web*. 178–186.
- [11] Tenzing C Dolmans, Mannes Poel, Jan-Willem JR van't Klooster, et al. 2020. Perceived Mental Workload Classification Using Intermediate Fusion Multimodal Deep Learning. *Frontiers in human neuroscience* 14 (2020), 609096.
- [12] Yotam Eshel, Noam Cohen, Kira Radinsky, et al. 2017. Named Entity Disambiguation for Noisy Text. In *Proceedings of the 21st Conference on Computational Natural Language Learning*. 58–68.
- [13] Andrea Frome, Greg S Corrado, Jonathon Shlens, et al. 2013. DeViSE: A Deep Visual-semantic Embedding Model. In *Proceedings of the 26th International Conference on Neural Information Processing Systems*. 2121–2129.
- [14] Konrad Gadzicki, Razieh Khamsehashari, and Christoph Zetzsche. 2020. Early vs Late Fusion in Multimodal Convolutional Neural Networks. In *Proceedings of the 23th IEEE International Conference on Information Fusion*. 1–6.
- [15] Jingru Gan, Jinchang Luo, Haiwei Wang, et al. 2021. Multimodal Entity Linking: A New Dataset and A Baseline. In *Proceedings of the 29th ACM International Conference on Multimedia*.
- [16] Octavian-Eugen Ganeva and Thomas Hofmann. 2017. Deep Joint Entity Disambiguation with Local Neural Attention. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*. 2619–2629.
- [17] Faegheh Hasibi, Krisztian Balog, and Svein Erik Bratsberg. 2016. Exploiting entity linking in queries for entity retrieval. In *Proceedings of the 2016 ACM international conference on the theory of information retrieval*. 209–218.
- [18] Kaiming He, Xiangyu Zhang, Shaoqing Ren, et al. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.
- [19] Johannes Hoffart, Stephan Seufert, Dat Ba Nguyen, et al. 2012. KORE: Keyphrase Overlap Relatedness for Entity Disambiguation. In *Proceedings of the 21st ACM International Conference on Information and Knowledge Management*. 545–554.
- [20] Johannes Hoffart, Mohamed Amir Yosef, Ilaria Bordino, et al. 2011. Robust Disambiguation of Named Entities in Text. In *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*. 782–792.
- [21] Sayali Kulkarni, Amit Singh, Ganesh Ramakrishnan, et al. 2009. Collective Annotation of Wikipedia Entities in Web Text. In *Proceedings of the 15th ACM SIGKDD The International Conference on Knowledge Discovery and Data Mining*. 457–466.
- [22] Phong Le and Ivan Titov. 2018. Improving Entity Linking by Modeling Latent Relations between Mentions. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*. 1595–1604.
- [23] I. Loshchilov and F. Hutter. 2017. Fixing Weight Decay Regularization in Adam. *CoRR* (2017). <http://arxiv.org/abs/1711.05101>
- [24] Jiasen Lu, Jianwei Yang, Dhruv Batra, et al. 2016. Hierarchical Question-image Co-attention for Visual Question Answering. In *Proceedings of the 30th International Conference on Neural Information Processing Systems*. 289–297.
- [25] Rada Mihalcea and Andras Csoma. 2007. Wikify! Linking Documents to Encyclopedic Knowledge. In *Proceedings of the 16th ACM conference on Conference on Information and Knowledge Management*. 233–242.
- [26] David Milne and Ian H Witten. 2008. Learning to Link with Wikipedia. In *Proceedings of the 17th ACM Conference on Information and Knowledge Management*. 509–518.
- [27] Volodymyr Mnih, Nicolas Heess, Alex Graves, et al. 2014. Recurrent Models of Visual Attention. In *Proceedings of the 27th International Conference on Neural Information Processing Systems*. 2204–2212.
- [28] Seungwhan Moon, Leonardo Neves, and Vitor Carvalho. 2018. Multimodal Named Entity Disambiguation for Noisy Social Media Posts. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*. 2000–2008.
- [29] Eric Müller-Budack, Jonas Theiner, Sebastian Diering, et al. 2021. Multimodal News Analytics Using Measures of Cross-modal Entity and Context Consistency. *International Journal of Multimedia Information Retrieval* 10 (2021), 111–125.
- [30] Jiquan Ngiam, Aditya Khosla, Mingyu Kim, et al. 2011. Multimodal Deep Learning. In *Proceedings of the 28th International Conference on Machine Learning*. 689–696.
- [31] Jeffrey Pennington, Richard Socher, and Christopher D Manning. 2014. Glove: Global Vectors for Word Representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*. 1532–1543.
- [32] Chenwei Ran, Wei Shen, and Jianyong Wang. 2018. An Attention Factor Graph Model for Tweet Entity Linking. In *Proceedings of the 2018 World Wide Web Conference*. 1135–1144.
- [33] Michael Röder, Ricardo Usbeck, Sebastian Hellmann, et al. 2014. N³-A Collection of Datasets for Named Entity Recognition and Disambiguation in the NLP Interchange Format. In *Proceedings of the 9th International Conference on Language Resources and Evaluation*. 3529–3533.
- [34] Sanket Shah, Anand Mishra, Naganand Yadati, et al. 2019. Kvqa: Knowledge-aware Visual Question Answering. In *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*. 8876–8884.
- [35] Wei Shen, Jianyong Wang, and Jiawei Han. 2015. Entity Linking with a Knowledge Base: Issues, Techniques, and Solutions. *IEEE Transactions on Knowledge and Data Engineering* 27 (2015), 443–460.
- [36] Cees GM Snoek, Marcel Worring, and Arnold WM Smeulders. 2005. Early Versus Late Fusion in Semantic Video Analysis. In *Proceedings of the 13th Annual ACM International Conference on Multimedia*. 399–402.
- [37] Nitish Srivastava and Ruslan Salakhutdinov. 2012. Multimodal Learning with Deep Boltzmann Machines. In *Proceedings of the 25th International Conference on Neural Information Processing Systems*. 2222–2230.
- [38] Yao-Hung Hubert Tsai, Shaojie Bai, Paul Pu Liang, et al. 2019. Multimodal Transformer for Unaligned Multimodal Language Sequences. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. 6558.
- [39] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. 2017. Attention is All you Need. In *Proceedings of the 31st Neural Information Processing Systems*. 5998–6008.
- [40] Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A Free Collaborative Knowledgebase. *Commun. ACM* 57, 10 (2014), 78–85.
- [41] Meng Wang, Haoen Wang, Guilin Qi, et al. 2020. Richpedia: A Large-Scale, Comprehensive Multi-Modal Knowledge Graph. *Big Data Research* 22 (2020), 100159.
- [42] Yansen Wang, Ying Shen, Zhun Liu, et al. 2019. Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors. In *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*. 7216–7223.
- [43] Jennifer Williams, Ramona Comanescu, Oana Radu, et al. 2018. Dnn Multimodal Fusion Techniques for Predicting Video Sentiment. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*. 64–72.
- [44] Ledell Wu, Fabio Petroni, Martin Josifoski, Sebastian Riedel, and Luke Zettlemoyer. 2020. Scalable Zero-shot Entity Linking with Dense Entity Retrieval. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 6397–6407.
- [45] Zhiwei Wu, Changmeng Zheng, Yi Cai, et al. 2020. Multimodal Representation with Embedded Visual Guiding Objects for Named Entity Recognition in Social Media Posts. In *Proceedings of the 28th ACM International Conference on Multimedia*. 1038–1046.
- [46] Ming-Wei Yih, Wen-tau an Chang, Xiaodong He, et al. 2015. Semantic Parsing via Staged Query Graph Generation: Question Answering with Knowledge Base. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics*. 1321–1331.
- [47] Li Zhang, Yulong Zhang, Zhixu Li, Qiang Yang, et al. 2021. Attention-based Multimodal Entity Linking with High-Quality Images. In *Proceedings of the 26th International Conference on Database Systems for Advanced Applications*. 533–548.
- [48] Qi Zhang, Jinlan Fu, Xiaoyu Liu, et al. 2018. Adaptive Co-attention Network for Named Entity Recognition in Tweets. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*. 5674–5681.